ENGINEERING PORTFOLIO · 2026

Chehak Aggarwal

I design mechanisms for robots and vehicles that work where people can't: Mars-analog terrain, the Arctic ocean, underwater.

SCHOOL	UC San Diego, Jacobs School of Eng.
DEGREE	B.S. Aerospace Engineering
SPECIALTY	Astrodynamics & Space Applications
GRAD	June 2029 · GPA 3.9
AWARD	Triton Scholarship ($80,000)

CAD	Onshape · CATIA V5 · Fusion 360
ANALYSIS	FEA · MSC ADAMS · kinematics
FAB	3D printing · laser cutting · lathe
CODE	Python · MATLAB · GitLab
SEEKING	Mech. design & robotics roles

[Hero image: rover arm or JELIX render]
images/rover-full.png

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DWG 01	JELIX: jellyfish-inspired underwater propulsion	Design · simulation
DWG 02	Mars rover arm: science claw & electronics enclosure	Built · used at URC
DWG 03	Arctic surface vessel: frame, LiDAR mount, pool tests	Co-design · tested
DWG 04	Lead-screw gripper (adapted) · other work	CAD · motion study

DWG 01 · RESEARCH, GENEX SPACE · SOLO DESIGN & ANALYSIS · JUN 2024 – JUL 2025

JELIX: jellyfish-inspired underwater propulsion

A propeller-free propulsion mechanism for small underwater vehicles. A cam–follower drive with spring return snaps four ribs inward to push out a jet of water, copying the pulse of *Cotylorhiza tuberculata*. I designed the cam in CATIA V5 and simulated it in MSC ADAMS, supervised by Theja K V.

65 mm	80 mm	0–40°	12 s
rib-tip travel X	rib-tip travel Y	rib stroke	cycle at 5 rpm

[Labeled cutaway]
images/jelix-labeled.png

Cutaway: motor and gear train drive the cam; the ribs are pivoted followers.

HOW IT WORKS

- **Expand:** the cam slowly raises the ribs and compresses the springs.
- **Contract:** the cam profile drops, the springs snap the ribs back, and a jet of water pushes the vehicle.

ANALYSIS

- Cam profile designed at 10° increments in CATIA V5.
- MSC ADAMS multibody simulation at 5 rpm over 24 s.
- Free-body model: F_y on each rib produces thrust.

WHY A CAM

One motor, low voltage, simple to build and maintain. Smart materials need more complex electronics.

[ADAMS plots]
images/jelix-adams-plots.png

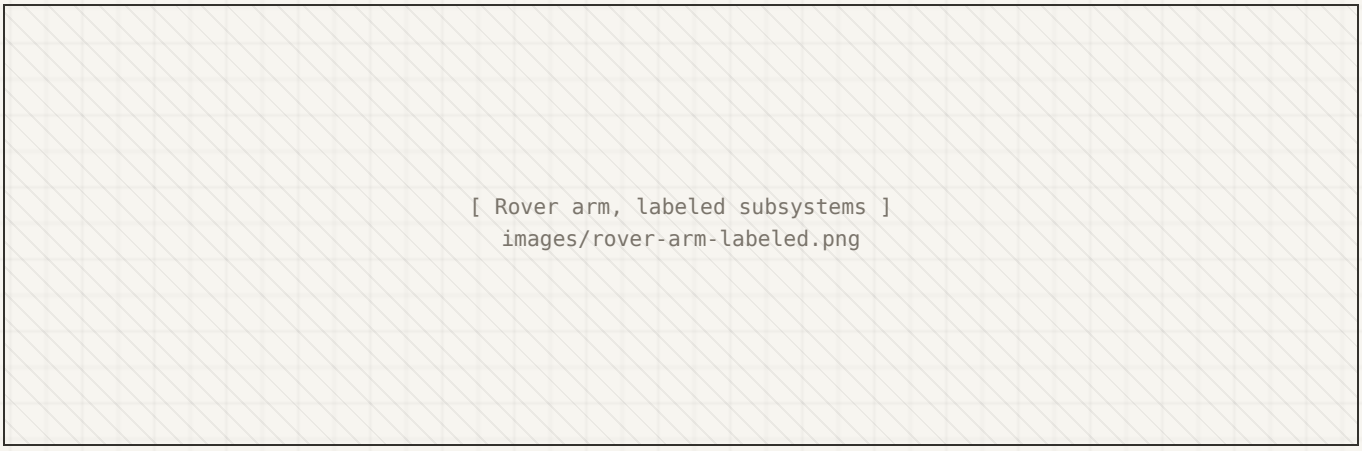
[Mechanism phases (a)–(e)]
images/jelix-mechanism.png

Status: simulation only, not yet built. Next: model the springs and the hydrodynamics, then print a one-rib rig and compare it against simulation.

PROJECT	TOOLS	DWG	SHEET
JELIX	CATIA V5 · MSC ADAMS	01	2 / 5

DWG 02 · YONDER DYNAMICS (UNIVERSITY ROVER CHALLENGE) · MECHANICAL TEAM · NOV 2025 — PRESENT

Mars rover arm: science claw & electronics enclosure



1 · SCIENCE CLAW: DESIGNED, BUILT, USED

The old end effector was hard to swap. I designed a **four-bar linkage** claw driven by **two servos**, with **replaceable fingers** and a direct bolt-on interface to the hand.

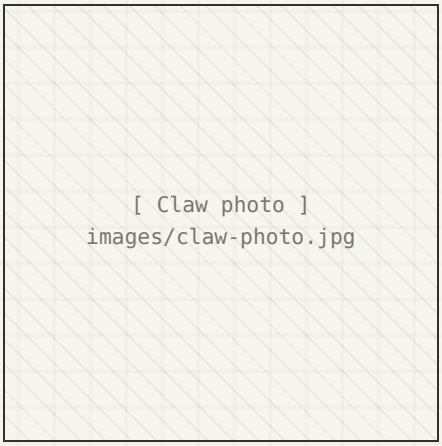
> **Result:** collected geological samples at URC 2026.

2 · FOREARM ENCLOSURE: REDESIGNED

The new ODrive and fuse board didn't fit; cramped wiring pressed on the fuse board and made servicing hard.

Length	178 → 216 mm (+21%)
Walls	3 mm
Cable exit	Enlarged for bend radius
Status	PLA prototype; arm redesigned







3 · ELBOW FEA: ANALYSIS SUPPORT

Ran static von Mises FEA on a teammate's elbow-joint design and found stress concentrations at the mounting interfaces.



PROJECT	TOOLS	DWG	SHEET
ROVER ARM	Onshape · FEA · 3D printing	02	3 / 5

DWG 03 · YONDER DEEP · MECHANICAL TEAM · NOV 2025 — PRESENT

Arctic surface vessel: frame, LiDAR mount, pool tests

An autonomous surface vessel that will collect underwater acoustic data with a hydrophone in the Arctic Ocean. I work on its structure and sensor integration, and helped run its first pool tests.



1 · PONTOON FRAME: CO-DESIGNED

Material	6061-T6 aluminum
Joining	Welded + bolted
Draft target	≈ 5 cm
Driver	Modular, field-serviceable

Co-designed with one other mechanical member, through team-wide review rounds.

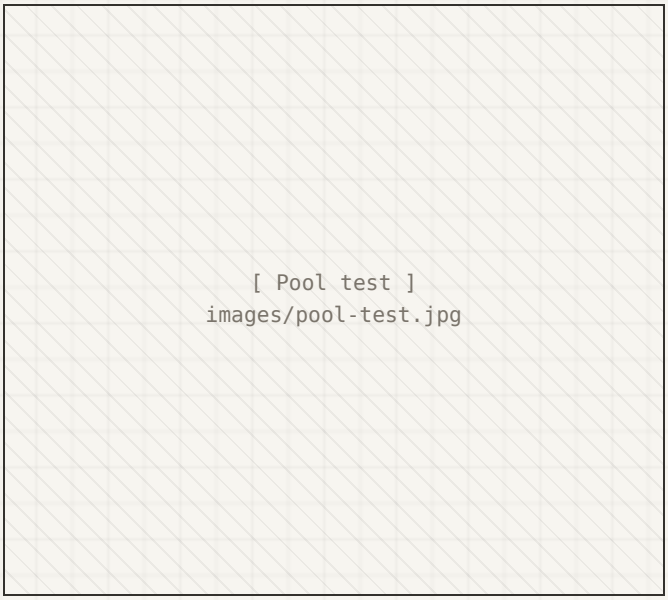
2 · LIDAR MOUNT: DESIGNED

Clean-sheet mount in Fusion 360 for stable sensing and a clear field of view in a marine environment.

3 · POOL TESTS: SUPPORTED

- Preliminary rig, X-configuration thrusters.
- Checked buoyancy, power draw, and hydrophone noise (no interference).
- Test 1: one motor failed. Test 2: out-and-back run completed.

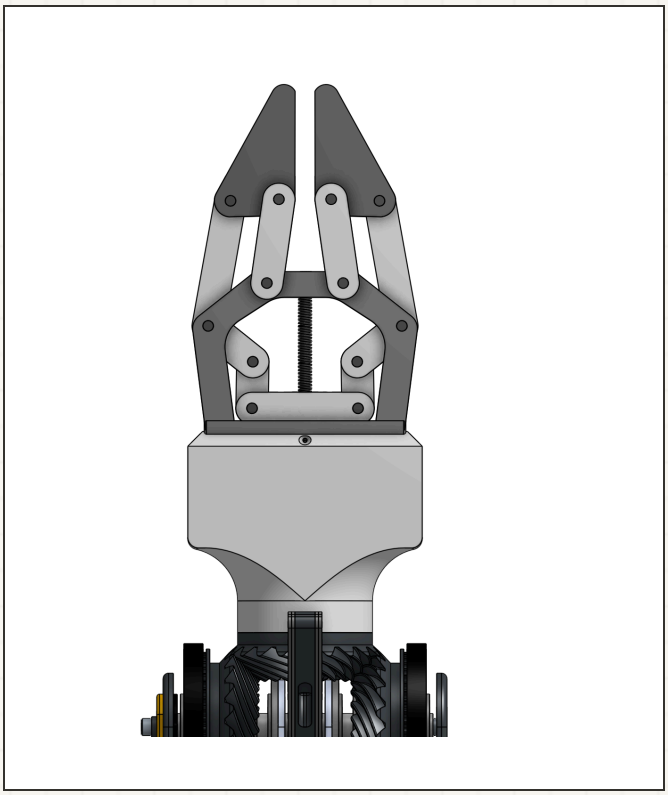




PROJECT	TOOLS	DWG	SHEET
ARCTIC USV	Fusion 360 · 3D printing	03	4 / 5

DWG 04 · ADAPTED DESIGN · ONSHAPE · OCT 2026

Lead-screw robotic gripper



A central lead screw drives a carriage that pushes mirrored four-bar linkages, swinging two fingertips together. I adapted an existing design and modeled it with a motion study in Onshape as a Yonder Dynamics design task.

WHY A LEAD SCREW

- High force from a small motor.
- Can be self-locking, so it holds without power.
- Trade-off: slower to close than a direct servo.

> Status: CAD and motion study; not manufactured. Motion video on the portfolio site.

Other work

HuLC '26	NASA Human Lander Challenge: co-researched T-PANS, a tunable viscoelastic damping concept for life-support noise on crewed lunar missions; wrote the Design Rationale & Value section.	AIAA UCSD
AIAA	Project Coordinator: recruited 32 students from 57+ applicants onto four NASA competition teams (HuLC, RASC-AL, Blue Skies, ORBITS).	Jun 2026 –
INTERN	Big Data Rhino: Python analysis of satellite data for proof-of-concept products; taught AI classes to 48 participants.	Summer 2026
ALSO	International Aerospace Olympiad, 2nd rank · competitive boxer (South Asian Games gold) · digital artist (64k followers)	–